

Equity-Aware Ambulance Outpost Siting in Sariyer

INDR 486/586 Term Project

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Problem Motivation

- Emergency response access depends strongly on facility location.
- Sarıyer has spatially dispersed neighborhoods and uneven accessibility.
- The project asks where ambulance outposts should be located to improve access.
- We evaluate both efficiency and equity.

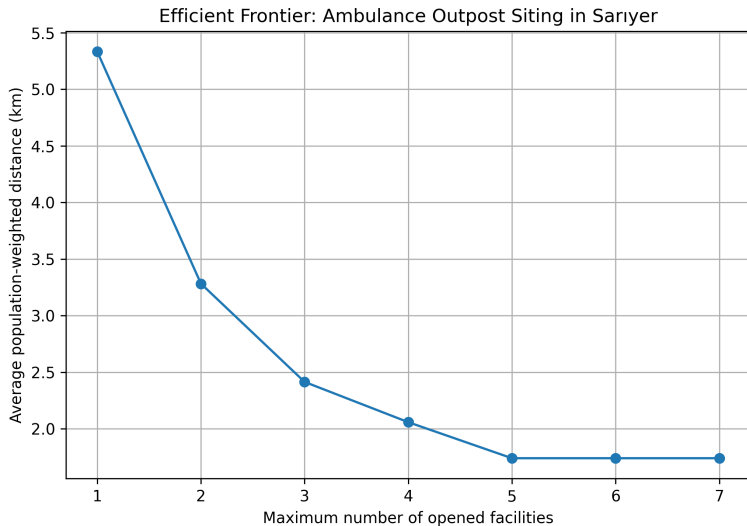
- 10 selected demand neighborhoods.
- Population is used as demand weight.
- 7 candidate public facilities.
- Driving distances are represented by a seed distance matrix.
- Optional final version can refresh distances using OSRM or openrouteservice.

Optimization Model

- Binary variable y_j : whether facility j is opened.
- Binary variable x_{ij} : whether neighborhood i is assigned to facility j .
- Objective: minimize population-weighted average distance.
- Constraint: each demand point assigned to one open facility.
- Constraint: at most P facilities can be opened.

- Maximum assigned distance.
- Weighted 90th percentile distance.
- Weighted coefficient of variation.
- Weighted Gini coefficient.

Efficient Frontier



Recommended Solution

- Recommended number of outposts: 4.
- Opened facilities: F2, F3, F6, F7.
- Average weighted distance: approximately 2.06 km.
- Maximum assigned distance: approximately 3.6 km.
- This solution balances accessibility, equity, and implementation cost.

Assignments

Neighborhood	Assigned Facility
Ayazaga	F3
Zekeriyakoy	F7
Resitpasa	F3
Tarabya	F2
Fatih Sultan Mehmet	F2
Istinye	F2
Yenikoy	F2
Ferahevler	F2
Maden	F6
Merkez	F6

Limitations

- Population is only a proxy for EMS demand.
- Real ambulance demand would require incident-level call data.
- Traffic and time-of-day effects are not modeled.
- The seed matrix should be refreshed with live road-network distances if possible.

Conclusion

- The model provides a reproducible district-scale planning framework.
- Four facilities appear sufficient under the seed instance.
- Equity metrics show whether any neighborhoods remain poorly served.
- The framework can be extended with real EMS demand and travel-time data.